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Backup & Recovery

Legacy backup tool

Native Workspace Key Backup and Recovery with Python Script

Note

This article covers the Python script (Legacy) used for key creation and disaster recovery. However, we strongly recommend that you use our [Native Recovery Utility tool](#).

We are deprecating the Python script in the future. The new Recovery Utility tool provides a simpler user experience for withdrawals, derivation, and key backup creation. If you are unable to use macOS or Ubuntu (M1/M2 chips) for your offline machine to perform the recovery tool backup, follow the guidelines below.

Overview

The Workspace Key Backup and Recovery process ensures that the two cloud key shares of your workspace's full private key are not stored only in Fireblocks' separate data centers. By not storing those two key shares only on Fireblocks' servers, you maintain full control of your workspace's full private key if you need to recover your assets from your Fireblocks vault.

Important

You must run a Workspace Key Backup for your workspace's full private key.

When you use Fireblocks MPC wallets, you keep direct custody of your assets. Therefore, as your own custodian, you are responsible for ensuring that you have independent access to your private keys. By backing up your workspace's full private key, you maintain access to your assets even if Fireblocks service is disrupted.

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Workspace Key Backup and Recovery processes

Workspace Key Backup and Recovery consists of two processes:

1. Setting up and testing in a testnet workspace.
2. Executing your Disaster Recovery (DR) plan in your production workspace.

The next sections show two ways you obtain your Workspace Key Backup package, maintain it, and perform tests in your testnet workspace and then your production workspace.

Workspace Key Recovery in a testnet workspace

Important

The process of backing up your production workspace's full private key can place your assets at risk. For that reason, we do not recommend running tests using your production workspace keys. You should always test the Workspace Key Backup and Recovery process in a testnet workspace.

Fireblocks allows you to run Workspace Key Recovery tests in a testnet workspace. The test consists of the same steps as the production Workspace Key Backup process. However, it allows you to familiarize yourself with the process without putting your signing keys at risk of exposure.

As a Fireblocks customer, you should have been given a testnet workspace. If you did not, contact your [Customer Success Manager](#) to obtain one to perform the Backup Simulation.

How to create your Workspace Key Backup (Python script) Legacy

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1. If you have a workspace that has been previously configured for recovery, your passphrase is already available. If they do not remember it, they can [reset it following these steps](#) before performing the backup.
2. Set up a secure, air-gapped machine for the offline backup. We recommended an Ubuntu 18.04 or newer operating system due to its minimal reliance on third-party libraries and dependencies, making it less susceptible to compromised attempts. However, any other operating system may also be suitable if the computer remains offline and does not go online again. Additionally, consider these security measures for your machine:
 - a. Make it accessible only to necessary authorized personnel.
 - b. Protect it with a very strong password.
 - c. Encrypt all partitions.
 - d. Store it in a physical safe when you are not using it.
3. We recommend you purchase additional dedicated hardware (e.g., a USB memory stick) to securely transfer files to and from your offline machine.
4. On your offline machine, go to our [Key Backup and Recovery Tool GitHub repo](#) and [follow the instructions](#) to install and run the Fireblocks Key Backup and Recovery Tool.
5. We recommend you use a physical safe to store a [separate keypair passphrase that you will create](#) during the recovery process and use it when you perform periodic backup verifications or an actual recovery.

The recommendations described above uses up to four offline devices for a basic setup or up to seven offline devices for an advanced setup.

Basic setup

For the basic setup, the following equipment is required:

- An offline machine that you store in a safe, to hold and run three key components:
 - Workspace Key Backup package provided to you by Fireblocks Support. It is described in detail in [Verifying your Workspace Key Backup package with Python Script](#).
 - Fireblocks Key Backup and Recovery Tool.
- A second offline machine or secure removable media (e.g., USB memory stick) for storing your recovery keypair, as detailed in step 2 below.
- Another offline machine, secure removable media (e.g., USB), or paper, for [storing your recovery passphrase](#).
- A secure USB for transferring your files to and from your offline machine.

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See memory section for the transfer in the advanced setup process.

Step 2: Generate your recovery keypair

The Fireblocks Key Backup and Recovery Tool uses the encryption standard AES-128, although you can use OpenSSL and use AES encryption levels AES-128, AES-196, and AES-256 instead of running it via the tool.

Your recovery keypair allows you to encrypt your two cloud key shares, and then decrypt them in an offline environment. To generate your recovery keypair, complete the steps below on your designated offline machine using command prompts.

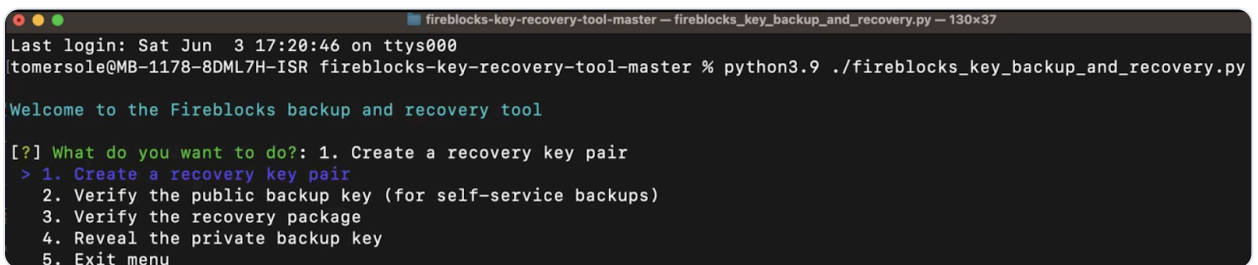
Note

If you are using a Windows machine, you must [first install OpenSSL](#) on the device.

1. Run the Fireblocks Key Backup and Recovery Tool:

```
./fireblocks_key_backup_and_recovery.py
```

2. Select **Create a recovery keypair** under the “What do you want to do?” prompt that appears, and follow the instructions.



```
fireblocks-key-recovery-tool-master — fireblocks_key_backup_and_recovery.py — 130x37
Last login: Sat Jun  3 17:20:46 on ttys000
tomersole@MB-1178-8DML7H-ISR fireblocks-key-recovery-tool-master % python3.9 ./fireblocks_key_backup_and_recovery.py

Welcome to the Fireblocks backup and recovery tool

[?] What do you want to do?: 1. Create a recovery key pair
> 1. Create a recovery key pair
  2. Verify the public backup key (for self-service backups)
  3. Verify the recovery package
  4. Reveal the private backup key
  5. Exit menu
```

Warning

You must memorize your keypair passphrase and keep a copy in a separate, secure place like a safe.

By the end of this process, the recovery keypair is saved under the directory from which you are running the tool.

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1. The Owner extracts the Public Key file to an online machine with access to your Console.
2. In the Fireblocks Console, the Owner goes to **Settings > General > In-house key backup**, then selects **Create backup**.

Extended public keys 

The public keys and private keys of all of wallets in this workspace are derived from the extended public keys and private keys. [Learn more](#)

Key backup

In-house key backup

You haven't backed up your keys. A backup is crucial to maintain full asset custodianship. [Learn more](#)

Create a key backup

The workspace owner must create a key backup.

Create backup

3. Select the public key file generated above > **Upload key**.

Create a key backup ×

Start the key backup process by uploading a public key file to encrypt your keys.
[Learn more](#)

Public key file (PEM)

 Drop your files here, or [browse](#)

 We recommend creating a key backup only after there is an admin quorum in place [Learn more](#)

[Cancel](#) [Upload key](#) 

4. Your workspace Owner receives a confirmation email stating that they and the Admin Quorum must now verify and approve the public recovery key on the Fireblocks mobile app.
5. Your Admin Quorum is notified to approve the key using the Fireblocks mobile app. If they do not approve it within 48 hours you must restart this process.
6. While the request is pending approval, selecting the yellow **Awaiting approval** badge shows the remaining approval requests and which Admins can still approve

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You have a key backup in progress.



Key backup

Awaiting approval ⓘ

The key backup
workspace ac
Fireblocks iOS

Required Authorizations

Requires approval by 1 of:

Juliet Harker

Mandatory

Cancel

3rd-party key backup

Have Coincover create an

Create a key backup w

cks.com). There are no other
locks Android version 1.0.52 or

- If you have not run the Fireblocks Key Backup and Recovery Tool yet, do so now by using the below command. Make sure you have completed the prerequisite setup steps before using the tool.

```
./f./fireblocks_key_backup_and_recovery.py
```

On the screen you see four options under the first prompt: "What do you want to do?"

```
[?] What do you want to do?: 2. Verify the public backup key (for self-service backups)
1. Create a recovery key pair
> 2. Verify the public backup key (for self-service backups)
3. Verify the recovery package
4. Reveal the private backup key
5. Exit menu
```

- Select option 2 - **Verify the public backup key (for self-service backups)**. (The purpose of this step is for the Admin Quorum to verify that the public key generated in the tool is the same as the one the Admin Quorum sees in their mobile app.)
- Enter the public key file name created above.
- Select the verification method: If the Admin Quorum performing the verification is near the offline machine, you can select option (a), otherwise select option (b):

a. If **Display a scannable public key QR code** is selected:

- A file named **pub_key_qr.png** is generated and opens automatically.

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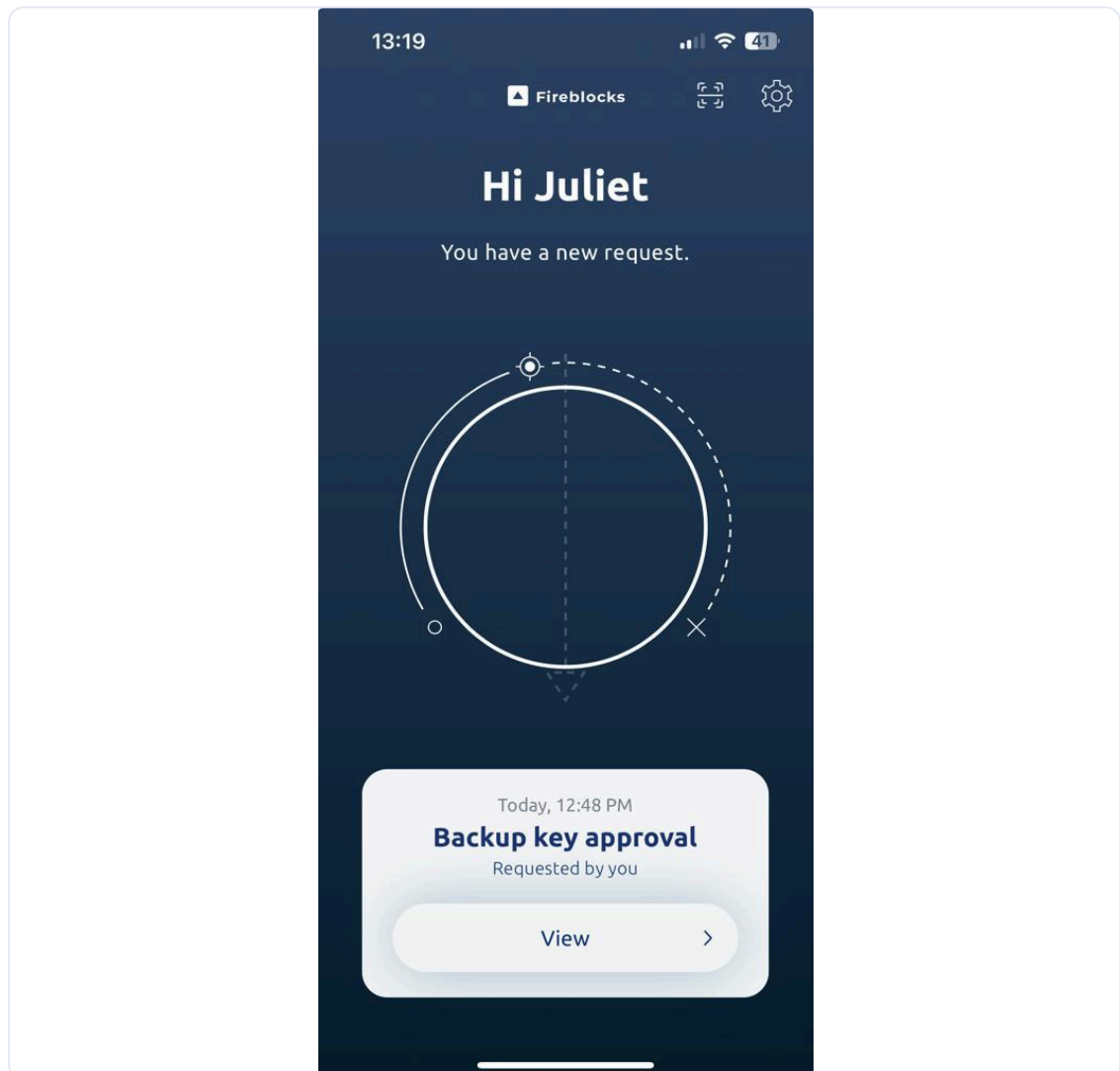
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- i. An eight-character short phrase is generated and shown in the tool.

```
[?] Choose the verification method: : 2. Obtain a public key short phrase
1. Display a scannable public key QR code
> 2. Obtain a public key short phrase
3. Exit menu

The short phrase is: D1Y3xP5H
```

11. Or your Admin Quorum collaborates with the workspace Owner to verify and approve the key backup by following the prompts on the Fireblocks mobile app:
- a. Select **View** > **Get Started** > **I'm ready to approve**.



- b. The Admin Quorum selects an approval method (either by scanning a QR code or inputting a short key) and informs the Owner. Since the Owner has access to the offline machine, they can proceed with the QR code option in

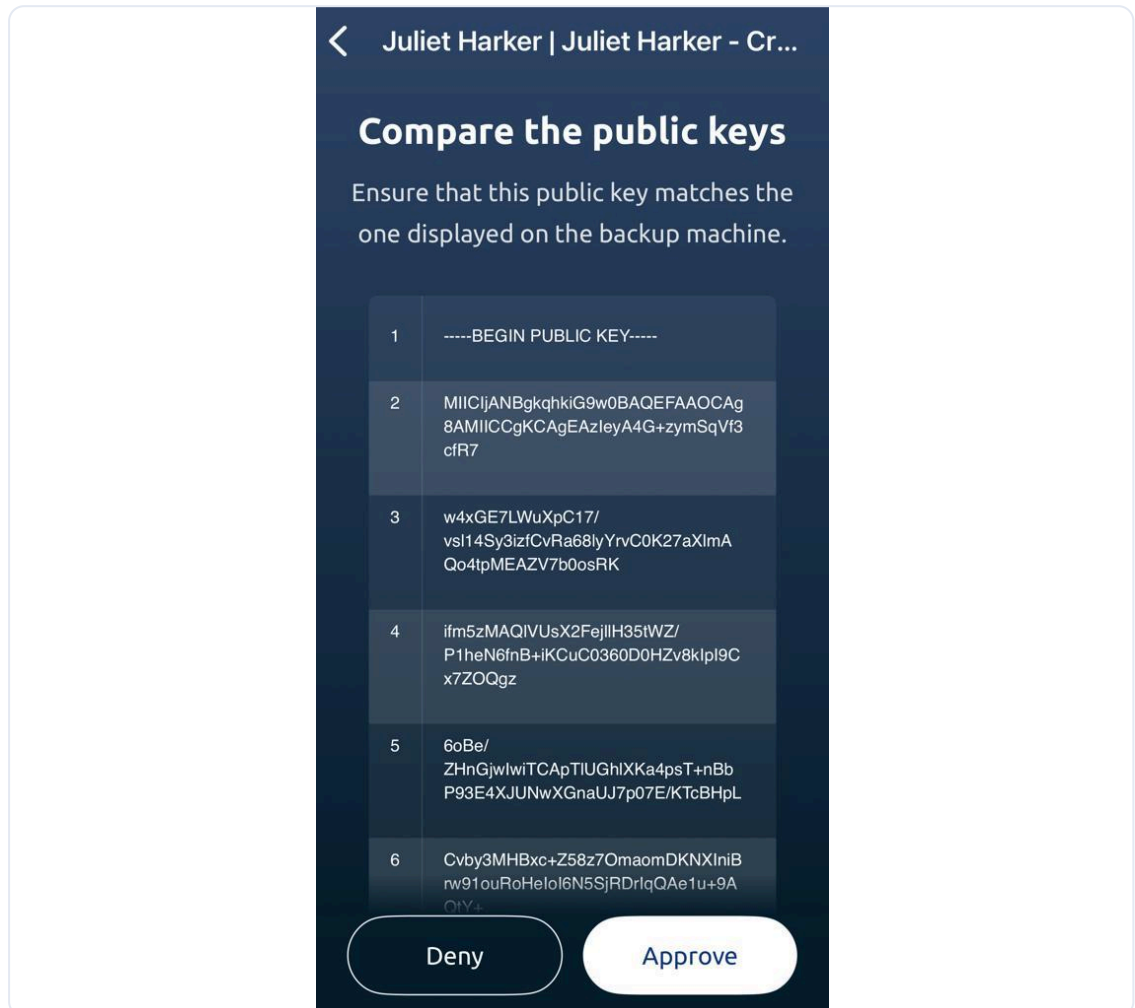
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the key has been verified.

e. The public key appears in the Fireblocks mobile app.



f. To view the public key on the offline machine, return to your Fireblocks Key Backup and Recovery Tool on the air-gapped machine and select **View the public key**.

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Compare the public key shown below to the one seen in the mobile app to verify the correct public key is used:

```
1. -----BEGIN PUBLIC KEY-----
2. MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAzIeyA4G+zymSqVf3cfR7
3. w4xGE7LWuXpC17/vsl14Sy3izfCvRa68lyYrvC0K27aXlmaQo4tpMEAZV7b0osRK
4. ifm5zMAQ1VUsX2FejllH35tWZ/P1heN6fnB+iKCuC0360D0HZv8kIpI9Cx7Z0Qgz
5. 6oBe/ZHnGjwIwiTCaPTLUGh1XKa4psT+nBbP93E4XJUNwXGnaUJ7p07E/KTcBHpL
6. Cvby3MHBxc+Z58z70maomDKNXIniBrw91ouRoHeIoI6N5SjRDrIqQAe1u+9AQtY+
7. EMNrYAYyGwcyWnPBgG4uSTM6JGmvXtw4LN4q80/rSDJnSNP4gavX3b9YyqU079T
8. Rt2/O/FUr1tC6Qh90ZEZQ7hPoig5MaYmspBJqjz9SBMFBvHAiETEF+0UB4zCaF5d
9. ZDyV5xoNkJSs1EQiYLA/Rr/Xi1eqpZoIKqQUPSur2DQkTe32SZy/h7/YmHZRYJR5
10. 0ojN1nimt125GLVG3r+U9f3po6cXBtr+YYzS6STQdHca15KLzGdCnxyGDRHkKyoP
11. CXE6wuD16NGR1CDu9sHUiWxskWjmWY+faFZkMZK9H3LvYF85o4ylqMikSUMj7WLe
12. bLg0uDJBHFYnZy8C801zaYIGgzvFwX0AHdzdPBNLr0Mf1rsStfsenGYJbub0NnKK
13. 4w08+Wx+dTg6eHdXhmeTbmcCAwEAAQ==
14. -----END PUBLIC KEY-----
```

Close

If the public keys match, select **Approve** on the Fireblocks mobile app. Otherwise, select **Deny**. This may mean the Owner accidentally modified or uploaded the wrong recovery public key at some point before it was submitted to your Console.

- g. The Admin Quorum and Owner enter their Fireblocks mobile app PIN codes and complete biometric authorization to approve the request.
- h. The Admin and Owner enter their Fireblocks app PIN codes and complete biometric authorization to approve the request.

12. In the Fireblocks Console, the backup status updates to **Awaiting completion**. If any Admin selected **Deny**, the backup status updates to **Denied**.

Key backup

In-house key backup

You have a key backup in progress.

Key backup Awaiting completion



The key backup was approved by Juliet Harker (jharker@fireblocks.com). There were no other workspace admins to approve the request. Download the package sent to your email address, follow the included instructions, and mark the backup as completed when you're done.

[Mark incomplete](#)

[Mark completed](#)

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Action required: Complete the key backup process

Juliet,

You're nearly done backing up your workspace keys.

To complete the final step:

- Download the key backup package.
- Safely move the package to the offline machine.
- Verify the key backup using the Fireblocks Key Backup and Recovery tool.



Handle with care!

Take strict security measures when working with the key backup package.

[View our security recommendations](#)

Link expires in 2 days.

[Download key backup package](#)

14. Once the included instructions in the email are carried out, mark the backup as completed in your Console.

Mark backup as completed?



By marking the key backup as completed, you confirm that:

- ☒ You downloaded the key backup package.
- ☒ The package was transferred to the offline machine.
- ☒ A sanity check of the key backup was completed at the offline machine using the Backup & Recovery tool. [Learn more](#)



As workspace owner, you are solely responsible for completing the key backup.

Cancel

Mark completed

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You've backed up your keys. We recommend verifying the key backup on the offline machine every 3 months. [Learn more](#)

Key backup 🕒 Completed



The key backup was approved by Juliet Harker (jharker@fireblocks.com). There were no other workspace admins to approve the request. It was marked as completed by Juliet Harker (jharker@fireblocks.com) on Jan 8, 12:48.

[Create another backup](#) [Show audit log](#)

16. You have now completed generating your in-house key backup package. Learn about verifying a recovery package [here](#).

Was this article helpful?

Yes

No

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